
Software Requirements Specification

for

Mobile Edge-Based Biometric Classroom Presence System

Version 1.0 approved

Prepared by the JetSet team

**Sorel Agbogla
Andres Moreira
Andrew Kim
Shayan Karki**

University of North Texas

February 2026

Table of Contents

Table of Contents	ii
1. Introduction.....	1
1.1 Purpose	1
1.2. Intended Audience and Reading Suggestions	2
1.3 Product Scope	2
1.4 References.....	2
2. Overall Description	2
2.1 Product Perspective	2
2.2 Product Functions	3
2.3 User Classes and Characteristics	3
2.4 Operating Environment	3
2.5 Design and Implementation Constraints.....	4
2.6 User Documentation	4
2.7 Assumptions and Dependencies	5
3. External Interface Requirements	5
3.1 User Interfaces	5
3.2 Hardware Interfaces.....	6
3.3 Software Interfaces	6
3.4 Communications Interfaces	6
4. System Features	7
4.1 System Feature 1	Error! Bookmark not defined.
4.2 System Feature 2 (and so on).....	7
5. Other Nonfunctional Requirements	Error! Bookmark not defined.
5.1 Performance Requirements.....	8
5.2 Safety Requirements.....	8
5.3 Security Requirements.....	8
5.4 Software Quality Attributes.....	8
5.5 Business Rules	9
6. Other Requirements	9
Appendix A: Glossary.....	9
Appendix B: Analysis Models	9
Appendix C: To Be Determined List.....	9

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and nonfunctional requirements for the Mobile Edge-Based Biometric Classroom Presence System, Version 1.0.

The system is an academic prototype designed to support classroom attendance verification using mobile-based facial recognition and geofencing. This document specifies system behavior, interfaces, constraints, and quality attributes required for implementation and evaluation within a single academic semester.

This SRS applies only to the prototype system and does not define institutional deployment or integration with production university systems.

1.2 Intended Audience and Reading Suggestions

This document is intended for:

- Development team members (mobile, backend, ML engineers)
- Project sponsor/instructor
- Testers and QA reviewers
- Future capstone teams maintaining or extending the project

Reading order recommendation:

1. Section 1 – Introduction
2. Section 2 – Overall Description
3. Section 4 – System Features
4. Section 5 – Nonfunctional Requirements
5. Appendices (for database and models)

1.3 Product Scope

Write a paragraph about the scope
In scope list functionalities
Out scope

The system provides a structured method for verifying classroom presence through mobile geofence detection and facial confirmation.

The product:

- Records entry and exit timestamps
- Stores attendance records locally
- Provides instructor review and manual override capability
- Exports data in a Canvas-aligned format

The product does not:

- Access Canvas databases directly
- Perform continuous tracking
- Store facial images permanently
- Operate as a surveillance system

1.4 References

- [1] Project Proposal and Requirements Draft, Internal Team Document, Version 1.0, February 2026
- [2] Canvas LMS Attendance Documentation (Public API/Schema references)
- [3] OpenCV Documentation, Version 4.x
- [4] Android Developer Documentation / iOS Developer Documentation

2. Overall Description

2.1 Product Perspective

The system is a standalone mobile and backend application. It does not replace Canvas but mirrors attendance structures for compatibility.

High-Level Architecture:

Mobile App

- Geofence Trigger
- Face Scan
- Local Biometric Processing
- Backend Attendance Storage
- Instructor Dashboard
- Export for Canvas Entry

The system operates independently of institutional databases.

2.2 Product Functions

The system shall:

- Detect classroom boundary entry using geofencing
- Enable face scanning only within the classroom boundary
- Verify student identity using facial recognition
- Record entry and exit timestamps
- Classify attendance status (Present, Late, Excused)
- Allow instructor manual overrides
- Export attendance records
- Prevent off-campus check-ins

2.3 User Classes and Characteristics

2.3.1 Administrators (High Priority)

- Technical oversight role
- Configure geofence boundaries
- Manage student enrollment
- Moderate technical expertise

2.3.2 Instructors (High Priority)

- Primary system users

- Review and adjust attendance records
- Moderate technical skills
- Use system regularly

2.3.3 Registered Students (High Priority)

- Use mobile app to check in
- Grant camera and location permissions
- Moderate smartphone literacy

2.4 Operating Environment

Mobile Platform:

- Android 13+ / iOS 16+

Backend Server:

- Ubuntu 22.04 LTS
- Python 3.11
- FastAPI (or equivalent framework)

Database:

- PostgreSQL 14

ML Library:

- OpenCV 4.x
- ArcFace

Instructor Interface:

- Web-based (React 18)

2.5 Design and Implementation Constraints

- No direct access to Canvas databases
- Facial recognition must execute locally (no cloud inference)

- Geofence must use device-native location APIs
- Prototype must be completed within one semester
- Must comply with university privacy standards

2.6 User Documentation

Delivered documentation:

- User Manual (PDF)
- Quick Start Guide for Students
- Instructor Guide for Attendance Review
- Setup Instructions for Geofence Configuration

Documentation will be written in clear, accessible language.

2.7 Assumptions and Dependencies

2.7.1 Assumptions

- A1. Students possess GPS-enabled smartphones.
- A2. Students grant camera and location permissions.
- A3. Classroom boundaries can be defined using GPS radius logic.

2.7.2 Dependencies

- D1. Mobile OS geolocation APIs
- D2. Open-source facial recognition libraries
- D3. Stable local backend server

3. External Interface Requirements

3.1 User Interfaces

Mobile App:

- Check-in screen

- Face scan confirmation screen
- Entry/exit status indicator
- Permission request prompts

Instructor Dashboard:

- Attendance table view
- Manual edit controls
- Export button
- Timestamp display

All interfaces follow standard UI conventions.

3.2 Hardware Interfaces

- Mobile device camera
- Mobile GPS/location sensors
- Local server hardware
- Standard classroom network (Wi-Fi)

3.3 Software Interfaces

- OpenCV 4.x (biometric processing)
- PostgreSQL 14 (database)
- Fast API backend services
- React 18 frontend
- Canvas schema reference (no direct API usage required)

Data exchanged internally uses JSON format.

3.4 Communications Interfaces

Communication occurs over HTTPS.

Client-server interactions use RESTful APIs.
All requests and responses use JSON format.

No continuous streaming or real-time messaging is required.
All data synchronization occurs via standard request-response cycles.

4. System Features

4.1 Classroom Geofence Activation

4.1.1 4.1.1 Description and Priority

Enables attendance verification only when a student enters a predefined classroom boundary.
Priority: High

4.1.2 4.1.2 Stimulus/Response Sequence

Stimulus: Student device enters classroom boundary
Response: Face scan functionality becomes available

Stimulus: Student exits boundary
Response: Exit timestamp recorded

4.1.3 4.1.3 Functional Requirements

REQ-GEO-1: The system shall define a configurable geofence radius.
REQ-GEO-2: The system shall enable check-in only within the geofence.
REQ-GEO-3: The system shall record entry timestamp.
REQ-GEO-4: The system shall record exit timestamp.

4.2 Facial Confirmation

4.2.1 Description and Priority

Confirms identity at check-in using local facial recognition.
Priority: High

4.2.2 Stimulus/Response Sequence

Stimulus: Student initiates face scan

Response: System verifies identity locally

Stimulus: Recognition failure

Response: Instructor override option remains available

4.2.3 Functional Requirements

REQ-FACE-1: The system shall process facial recognition locally.

REQ-FACE-2: The system shall not store raw facial images.

REQ-FACE-3: The system shall discard biometric data after verification.

REQ-FACE-4: The system shall notify the user of the verification result.

4.3 Attendance Record Management

Stores attendance and supports instructor adjustments.

Priority: High

REQ-ATT-1: The system shall store attendance records in a structured database.

REQ-ATT-2: The system shall classify attendance as Present, Late, or Excused.

REQ-ATT-3: The system shall allow instructor manual overrides.

REQ-ATT-4: The system shall export records in CSV format.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- Face recognition response time ≤ 3 seconds
- Geofence detection response ≤ 5 seconds
- System supports up to 50 concurrent students

5.2 Safety Requirements

- System shall not perform continuous tracking
- System shall disable check-in outside boundary
- Instructor authority supersedes automated decisions

5.3 Security Requirements

- All communications over HTTPS
- Role-based access control
- No persistent storage of facial images
- Encrypted database storage

5.4 Software Quality Attributes

- Reliability: $\geq 95\%$ successful recognition under normal lighting
- Maintainability: Modular architecture
- Usability: Check-in process ≤ 3 user actions
- Testability: Each module independently testable

5.5 Business Rules

- Only enrolled students may check in
- Instructor approval finalizes attendance
- Late classification threshold configurable
- Students may revoke consent

6. Other Requirements

- Database must include student ID, timestamp, status fields
- Data retention limited to academic semester
- System must support CSV export

Appendix A: Glossary

Geofence – A virtual geographic boundary

Local Processing – Execution on device or on-premise server

Canvas-Aligned Schema – Attendance format consistent with Canvas structure

Override – Instructor manual modification

Appendix B: Analysis Models

(Include in final document)

- System Architecture Diagram
- Database ER Diagram
- Use Case Diagram
- Data Flow Diagram

Database design is the most critical system component.

Appendix C: To Be Determined List

TBD-1: Final geofence radius value

TBD-2: Exact facial recognition library

TBD-3: Mobile platform final decision (Android vs cross-platform)